ETH zürich | Physikalisches Praktikum der ETH Zürich | Physics Lab

Name 1 | Datum:

Name 2 | Platz-Nr:

63 - RLC and Alternating Currents

Homework

1. Peak value and Effective Value

☐ I know the difference between peak value and effective value. The conversion between the two values is

_____

2. Filter

Filter from Fig. 6 .... : $\frac{U_{\text{out}}}{U_{\text{in}}} =$ _____

This is therefore a _____ -pass-filter

Filter from Fig. 6 .... : $\frac{U_{\text{out}}}{U_{\text{in}}} =$ _____

This is therefore a _____ -pass-filter

3. (voluntary) Difference RC- and LC-Filter

Please explain, which of the two filters has the steeper edge, and why!

_____

_____

_____

Part I – Frequency-dependent voltage divider and filter

1. Low pass filter

$C =$_____ $R =$_____ $U_{\text{in}} =$_____

f	
ω	
U_{out}	
φ	

Graphical representation of $\frac{U_{\text{out}}}{U_{\text{in}}}$ as function of $\log(f)$!

2. Cut-off frequency

From the plot: $\omega_{\text{cutoff}}^{\text{exp}} =$ _____

Theoretical prediction: $\omega_{\text{cutoff}}^{\text{theo}} =$ _____

3. Limiting cases

Discuss the shape of the curve, especially for the limiting cases $\omega \ll \omega_{\text{cutoff}}$ and $\omega \gg \omega_{\text{cutoff}}$. How do you explain the shape of the curve for the limiting cases?

_____

_____

_____

_____

4. Roll-Off

The roll-off is about _____ *[indicate units!]*

5. Phase shift between U_{in} and U_{out}

Plot the phase shift φ between U_{in} and U_{out} as function of $\log \omega$, ideally in the same plot as for task 1. The phase shift at the cutoff frequency is $\varphi(\omega_{\text{cutoff}}) =$_____.

Phase shift between current and voltage across a capacitor

The phase shift φ between current and voltage across a capacitor at the cutoff frequency ω_{cutoff} is _____.

The phase shifts for some other frequencies are:

Part II – RLC parallel circuit

1. Parallel RLC resonant circuit

C =_____ R =_____ L = _____ U_{in} =_____

From these values we get: ω_0^{theory} =_____

f	
ω	
U_{out}	
φ	

2. Graphical representation of U_{out} as function of ω

3. Quality factor Q of the resonant circuit

(*determine either using a fit and/or by graphical representation*

$Q_{\text{graphical}}$ = _____ ± _____

Q_{fit} = _____ ± _____ *(only D-PHYS)*

4. Current through a resonant circuit

C =_____ R =_____ L = _____ U_{in} =_____

f	
ω	
I_{R}	
φ	

Graphical representation of φ as function of ω.

5. (only D-PHYS) Resonance frequencies for various L and C

i) $C =$ _____ ; $L =$ _____ ; $\omega_0 =$ _____

ii) $C =$ _____ ; $L =$ _____ ; $\omega_0 =$ _____

iii) $C =$ _____ ; $L =$ _____ ; $\omega_0 =$ _____

iv) $C =$ _____ ; $L =$ _____ ; $\omega_0 =$ _____

v) $C =$ _____ ; $L =$ _____ ; $\omega_0 =$ _____

vi) $C =$ _____ ; $L =$ _____ ; $\omega_0 =$ _____

Graphical representation of ω_0 as *meaningful* function $f(L, C)$.

6. (only D-PHYS) Square and triangular signal

Supply the RLC circuit with a rectangular or triangular signal and change U_{in}. What do you observe?

keywords: Ring down, Fourier transform of the input signal

7. (voluntary) Currents in a parallel circuit

Measure the currents I_L and I_C with the multimeter. What do you observe? How large are the currents relative to the current across the resistor I_R??